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## 單選題 一題 2.5 分

## 1. In an ecosystem, energy

- A) cycles along with chemical nutrients.
- B) typically flows from consumers to producers to decomposers.
- C) typically flows from producers to a series of consumers.
- D) comes ultimately from bacteria.

## 2. Which of the following trace elements needed by humans is commonly added to table salt?

- A) iodine
- B) iron
- C) magnesium
- D) fluoride

## 3. Organic compounds

- A) always contain nitrogen.
- B) are synthesized only by animal cells.
- C) always contain carbon.
- D) always contain oxygen.

## 4. One centimeter = \_\_\_\_\_ millimeters.

- A) 0.01
- B) 0.10
- C) 10
- D) 100

## 5. The cholesterol associated with animal cell membranes

- A) is attached to membrane proteins and extends into the watery environment surrounding the cell.
- B) helps to keep phospholipids from being too close to one another.
- C) is an abnormality resulting from a diet high in cholesterol.
- D) helps solidify the membranes when the room temperature is below freezing.

## 6. How do cells capture the energy released by cellular respiration?

- A) They produce ATP.
- B) They produce glucose.
- C) They store energy in molecules of carbon dioxide.
- D) The energy is coupled to oxygen.

7. CO<sub>2</sub> enters and O<sub>2</sub> escapes from a leaf via

- A) stomata.
- B) thylakoids.
- C) grana.
- D) stroma.

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**8. Sister chromatids are**

- A) found immediately after a cell divides.
- B) joined together at a centromere.
- C) made only of DNA.
- D) unique to prokaryotes.

**9. The alleles of a gene are found at \_\_\_\_\_ chromosomes.**

- A) the same locus on nonhomologous
- B) different loci on homologous
- C) different loci on nonhomologous
- D) the same locus on homologous

**10. If one strand of DNA is CGGTAC, then the complementary strand would be**

- A) GCCTAG.
- B) GCCAUC.
- C) TAACGT.
- D) GCCATG.

**11. Which of the following enzymes catalyzes the elongation of a new DNA strand?**

- A) RNA polymerase
- B) DNA ligase
- C) reverse transcriptase
- D) DNA polymerase

**12. A gene operon consists of**

- A) a transcribed gene only.
- B) a promoter only.
- C) a regulatory gene only.
- D) transcribed genes, an operator, and a promoter.

**13. vaccine works by**

- A) inhibiting bacterial replication.
- B) stimulating the immune system.
- C) inhibiting viral replication.
- D) preventing the translation of mRNA.

**14. Restriction enzymes**

- A) facilitate nucleotide base pairing.
- B) cut DNA at specific sites.
- C) stop transcription and translation.
- D) bind together strands of DNA.

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15. The core theme of biology, which explains both the unity and diversity of life, is
- A) genetics.
  - B) ecology.
  - C) evolution.
  - D) metabolism.
16. Which of the following would a biologist describe as microevolution?
- A) the formation of new species
  - B) the extinction of species
  - C) dramatic biological changes, such as the origin of flight, within a taxon
  - D) a change in the gene pool of a population from one generation to the next
17. Which highly reactive gas was probably absent from the Earth's primitive atmosphere?
- A) methane
  - B) carbon dioxide
  - C) oxygen
  - D) water vapor
18. Prokaryotic cell walls function
- A) to promote flexibility and formation of pseudopodia.
  - B) to prevent the cell from bursting in a hypotonic environment.
  - C) to propel cells (locomotion).
  - D) as a site of metabolic reactions (photosynthesis and cellular respiration).
19. The ripened ovary of a flower, which is adapted to disperse seeds, is called a/an
- A) ovule.
  - B) casing.
  - C) fruit.
  - D) sporangium.
20. Which of the following animals displays radial symmetry?
- A) an earthworm
  - B) a sea anemone
  - C) a fish
  - D) a lobster
21. Which is characteristic of all vertebrates?
- A) a skull and a backbone consisting of vertebrae
  - B) a calcified (hard) skeleton and four legs
  - C) a hinged jaw
  - D) lungs or lung derivatives

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**22. Structure in the living world is organized into hierarchical levels. Which of the following correctly lists these from least inclusive to most inclusive?**

- A) cell, molecule, organ, organ system, tissue, organism
- B) molecule, cell, tissue, organ, organ system, organism
- C) molecule, cell, organ, organ system, tissue, organism
- D) cell, molecule, tissue, organ, organ system, organism

**23. Digestion is the**

- A) absorption of nutrients suspended in water.
- B) conversion of glycogen to glucose.
- C) breakdown of food into molecules small enough for the body to absorb.
- D) churning of food in the stomach and intestine.

**24. A waste product of cellular respiration is**

- A) electrons.
- B) hydrogen peroxide.
- C) carbon dioxide.
- D) glucose.

**25. In mammals, which of the following vessels transports oxygenated blood from the lung back to the heart?**

- A) pulmonary artery
- B) pulmonary vein
- C) superior vena cava
- D) coronary artery

**26. During gas exchange, body cells**

- A) take up CO<sub>2</sub> and release O<sub>2</sub>.
- B) release O<sub>2</sub> to red blood cells.
- C) increase in size to accommodate the reuptake of O<sub>2</sub>.
- D) release CO<sub>2</sub> and take up O<sub>2</sub>.

**27. Which of the following animals has a single circuit of blood flow and two heart chambers?**

- A) salamander
- B) salmon
- C) snake
- D) squirrel

**28. Natural killer cells**

- A) are phagocytes that attack and kill pathogenic microorganisms.
- B) attack virus-infected cells by releasing chemicals that lead to cell death.
- C) tag pathogenic microorganisms with antibodies.
- D) "eat" microorganisms that have been tagged with antibodies.

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**29. The two main functions of the lymphatic system are**

- A) coagulating blood and fighting infections.
- B) producing hormones that regulate the immune system and coagulating blood.
- C) producing hormones that regulate the immune system and fighting infections.
- D) returning tissue fluid to the circulatory system and fighting infections.

**30. Some complement proteins**

- A) induce antibody formation by phagocytic cells.
- B) help trigger the inflammatory response.
- C) are released by natural killer cells to attack cancer and virus-infected cells.
- D) replace T cells in the cell-mediated response.

**31. Animals that maintain internal body temperature using heat generated by their own metabolism are called**

- A) endotherms.
- B) ectotherms.
- C) reptiles.
- D) amphibians.

**32. Compared to the endocrine system, the nervous system has a**

- A) faster response, but the signal does not last as long.
- B) slower response, and a signal that does not last as long.
- C) faster response with a longer-lasting signal.
- D) slower response with a longer-lasting signal.

**33. Reproductive systems with external fertilization are most common in**

- A) terrestrial animals.
- B) populations with many more males than females.
- C) animals that are widely dispersed.
- D) aquatic animals.

**34. The functional unit of the nervous system is the**

- A) cell body.
- B) neuron.
- C) axon.
- D) synapse.

**35. In stimulus detection, sensory receptor cells convert a stimulus into a(n)**

- A) neurotransmitter.
- B) enzyme.
- C) electric signal.
- D) astigmatism.

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**36. Which structure constitutes part of the axial skeleton?**

- A) skull
- B) pelvic girdle
- C) pectoral girdle
- D) radius

**37. What is a characteristic of eudicots?**

- A) parallel leaf venation
- B) vascular bundles scattered throughout the stem
- C) a flower with six petals
- D) a taproot system

**38. The loss of water from the leaves of plants is**

- A) adhesion.
- B) cohesion.
- C) transpiration.
- D) osmosis.

**39. Plants grow toward light through the action of**

- A) hormones.
- B) nerves.
- C) phytochrome.
- D) chloroplasts.

**40. Which environmental factor has a direct effect on an organism's rate of water loss by evaporation?**

- A) insect biodiversity
- B) wind
- C) plant pollination rates
- D) algae abundance